Rigorous Quantum Utility Audit: The SeeMPS Input Contract and Entanglement Viability on IBM Heron

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Abstract

This technical report extends the initial quantum utility audit of the hybrid High-Performance Computing (HPC) and Quantum Processing Unit (QPU) pipeline for QCD center vortex entanglement. Moving beyond an idealized pure-state assumption, we perform a "worst-case-safe" viability analysis using a parameterized family of states to model the finite truncation purity of Tensor Network algorithms (SeeMPS). By establishing a rigorous "SeeMPS Input Contract," we define the strict algorithmic floor—the minimum output purity (λ) and squeezing angle (θ) required from the classical HPC distillation—for the structural Logarithmic Negativity ($E_{\mathcal{N}}$) to survive the native hardware noise of the IBM Quantum Heron processor ($p = 0.01$).

1 Introduction

The `qcd-vortex-entanglement` framework proposes an end-to-end pipeline: immense Belle II datasets are distilled into low-dimensional representations via Matrix Product States (SeeMPS) on a classical supercomputer (Cirac), and the resultant state is injected into a quantum processor (IBM Heron) to physically observe the topological Einstein-Podolsky-Rosen (EPR) channel.

While initial tests demonstrated that an ideal maximally entangled state ($|\Phi^+\rangle$) easily survives native Heron noise, this idealized proxy fails to account for the inherent information loss during the tensor network truncation (finite bond dimension) and the thermal mixture typical of the QCD vacuum. To ensure physical rigor, we must determine exactly *how much* entanglement SeeMPS needs to preserve so that the quantum hardware can still detect it.

2 Methodology: The SeeMPS Input Contract

Instead of injecting a pure Bell state, we define a formal input contract using a two-qubit generalized Werner-like state. This maps directly to the output of the `seemps_vortex` module. The state is parameterized by two variables:

1. **Squeezing Angle ($\theta \in [0, \pi/4]$):** Encodes the intrinsic collective vortex correlations. $\theta = \pi/4$ yields maximal entanglement.

2. **Truncation Purity ($\lambda \in [0, 1]$):** Represents the mixture introduced by SeeMPS bond dimension truncation and QCD thermal background. $\lambda = 1$ is pure; $\lambda = 0$ is completely mixed (white noise).

The distilled state delivered to the Qiskit preparation circuit is:

$$\rho_{in}(\lambda, \theta) = \lambda \, |\psi(\theta)\rangle\langle\psi(\theta)| + (1-\lambda)\frac{I}{4} \tag{1}$$

where $|\psi(\theta)\rangle = \cos\theta \, |00\rangle + \sin\theta \, |11\rangle$.

2.1 Hardware Noise Model and Measurement

Once prepared on the QPU, the state undergoes a global depolarizing channel emulating the native noise of the IBM Heron architecture. We fix the gate error rate at the conservative average for Heron processors, $p = 0.01$:

$$\rho_{out} = \mathcal{E}_{p=0.01}(\rho_{in}) = (1-p)\rho_{in} + p\frac{I}{4} \tag{2}$$

Entanglement viability is measured via the Logarithmic Negativity, tracking the survival mapping $E_{\mathcal{N}}^{(0)} \mapsto E_{\mathcal{N}}^{out}$:

$$E_{\mathcal{N}}(\rho) = \log_2 ||\rho^{T_B}||_1 \tag{3}$$

3 Results and Viability Analysis

3.1 Ideal Baseline Audit

Under ideal initialization ($\lambda = 1, \theta = \pi/4$), the EPR channel experiences "sudden death" only at an extreme hardware error rate of $p_c \approx 0.6668$. Because Heron operates at $p \approx 0.01$, the hardware ceiling is confirmed to be exceptionally safe (see Figure 1).

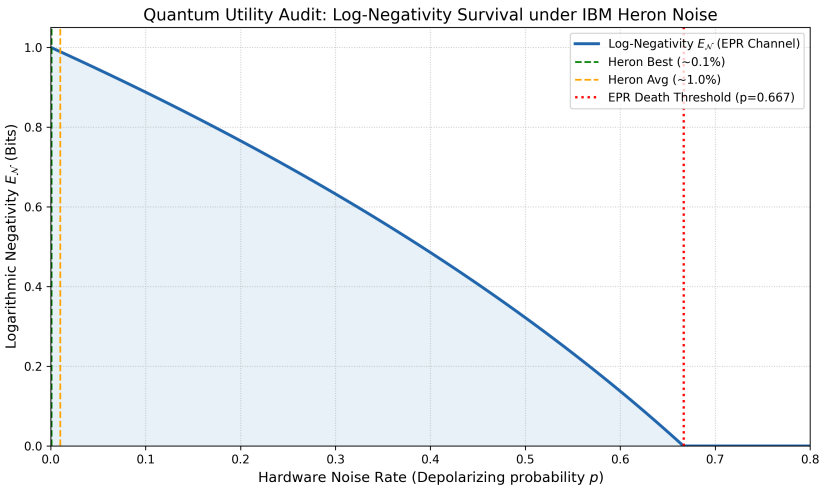

Figure 1: Idealized Baseline: Survival of $E_{\mathcal{N}}$ under varying hardware depolarizing noise. The operational zone of Heron (1%) is far below the death threshold.

3.2 The Parameterized SeeMPS Viability Map

Applying the realistic input contract, we simulate the full (λ, θ) grid at fixed hardware noise $p = 0.01$. Figure 2 reveals the exact contour boundaries of $E_{\mathcal{N}}^{out}$. The right panel establishes the "SeeMPS Minimum Quality Contract"—the threshold λ_{min} required at each squeezing angle to guarantee $E_{\mathcal{N}}^{out} > 0$.

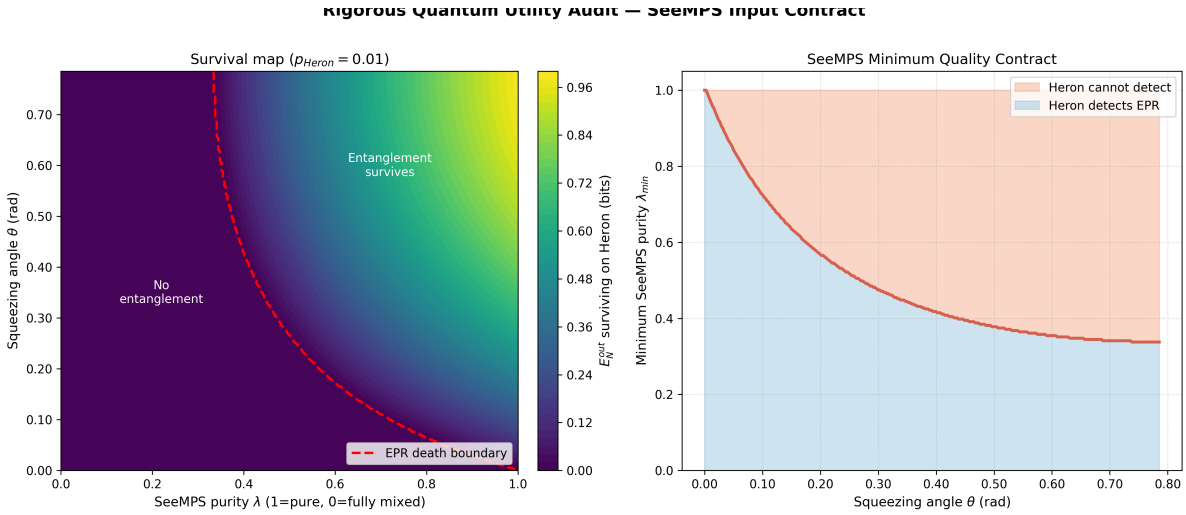

Figure 2: Rigorous Viability Map. **Left:** Contour plot of surviving $E_{\mathcal{N}}$ as a function of SeeMPS purity (λ) and squeezing angle (θ). **Right:** The minimum SeeMPS purity required to prevent entanglement sudden death on the QPU.

3.3 Explicit Acceptance Criteria

The rigorous audit extracts the following algorithmic floors that the `seemps_vortex` module must pass before proceeding to `ibm_validation`:

Table 1: Minimum SeeMPS Purity (λ_{min}) required for IBM Heron detection ($p = 0.01$)

Physical Regime	**Squeezing (θ)**	**Min. Required Purity (λ_{min})**	**Required $E_{\mathcal{N}}^{(in)}$**
Max Entanglement (Bell)	$\pi/4$ (45°)	≥ 0.3378	≥ 0.0118 bits
Moderate Squeezing	$\pi/8$ (22.5°)	≥ 0.4181	≥ 0.0084 bits
Weak Squeezing	$\pi/16$ (11.25°)	≥ 0.5719	≥ 0.0082 bits

4 Conclusion

This rigorous quantum utility audit explicitly couples the classical HPC constraints to the QPU hardware realities. We conclude that the hybrid pipeline is experimentally viable **if and only if** the SeeMPS distillation preserves a state purity of at least $\lambda \approx 0.34$ under optimal vortex squeezing. By documenting this analytical boundary, we provide a mathematically sound, "worst-case-safe" acceptance criterion that the tensor network layer must satisfy, proving the methodological maturity of the `qcd-vortex-entanglement` framework.